GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

# Interior and Exterior Angles of a Triangle

*Independent Set 6.T2* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Two angles of a triangle are 63° and 79°. Find the third angle.

***38°***

*180 − 142.*

**2.** The angle between the two equal sides of an isosceles triangle is 40°. Find each base angle.

***70°***

*(180 − 40) ÷ 2.*

**3.** A triangle's remote interior angles are 46° and 58°. Find the exterior angle.

***104°***

*46 + 58.*

**4.** An exterior angle of a triangle is 140°. One remote interior angle is 62°. Find the other.

***78°***

*140 − 62.*

**5.** A triangle's angles are $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$. Find each angle.

***40°, 60° and 80°***

*9x = 180, x = 20.*

**6.** An exterior angle of a triangle measures $(4x+10)^\circ$; its remote interior angles measure $50^\circ$ and $(2x)^\circ$. Write an equation and find x.

***4x + 10 = 50 + 2x; x = 20***

*2x = 40.*